MILITARY SPECIFICATION

HEATER, BITUMEN

This Amendment forms a part of Military Specification MIL-H-395F, dated 28 January 1971, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 1

2.1, under SPECIFICATIONS, Federal, add:
"ZZ-I-550 - Inner Tube, Pneumatic Tire."

Page 2

2.1, under SPECIFICATIONS, Military, add:
"MIL-T-12459 - Tire, Pneumatic: For Military Ground Vehicles.
MIL-C-13516/1 - Circuit Breakers; 28 Volt DC, Automatic, Waterproof, 15 to 30 Amp.
MIL-C-24066/2 - Clip, Solid, Spring Tension."

2.1, under STANDARDS, Military, add:
"MIL-STD-1474 - Noise Limits for Army Material."
Delete: "MS17160 - Clip, Spring Tension (for Cylindrical Electronic Component.)"

Page 3

2.1, under STANDARDS, Military, delete:
"MS35388 - Tire, Pneumatic: Passenger Car, Light Truck, Truck and Bus.
MS35392 - Inner Tube: Pneumatic Tire.
MS39062 - Circuit Breakers: Thermal, 30 Volt DC, Automatic, Waterproof, 15 to 30 Amp."

Change title of MS35643 to read "Gasket: Synthetic Rubber, Cap, Fuel Tank."

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2.2, under Other publications, SOCIETY OF AUTOMOTIVE ENGINEERS, delete: "HS-39-Construction and Industrial Machines Handbook Supplement."

Page 5

2.2, under Other publications, AMERICAN WELDING SOCIETY (AWS), change AWS to read: "2501 Northwest Seventh Street, Miami, FL 33125."

Page 6

2.2, under Other publications, delete:

"DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE".

Add:

"ASSOCIATION OF AMERICAN RAILROADS

Rules Governing the Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)"

Page 7

3.2, delete and substitute:

"3.2 First article (preproduction model). The supplier shall furnish a heater for examination and test within the time frame specified (see 6.2), to prove prior to starting production, that his production methods and choice of design detail will produce heaters that comply with the requirements of this specification. Examinations and tests shall be as specified in section 4 and unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the production examinations and tests, as specified (see 6.2)."

3.2.1, delete in its entirety.

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After 3.5, add:

"3.5.1 Human factors. The characteristics of the heater shall provide for operation by personnel ranging from the small man clothed, through the large man clothed in accordance with SAE J833.

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3.5.2 Noise level. When subjected to the test specified in 4.6.2.12, the sound level of the heater shall conform to MIL-STD-1474, Category C plus 5dB, for steady-state noise. Hearing protectors shall not be furnished with the heater. When applicable, hazard sign shall state 'CAUTION, Hearing Protection Required,' and shall be as specified in 3.30."

Page 13

3.15.2, line 9, delete the last sentence entirely: "The engines shall conform to . . . . . with Part 85, as applicable."

3.15.2.1, lines 5 and 6 delete "of the engine manufacturer's published maximum horse power rating," and substitute "of the engine horsepower rating, as determined in 4.3.1,".

Page 14

3.16.1, line 2, delete "55" and substitute "45".

3.16.3, line 5, delete "20 percent" and substitute "10 percent."

Page 24

3.19, in subparagraph (c), delete "MS35388-16" and substitute "MIL-T-12459, class cc, style a;" delete "MS35392-13" and substitute "ZZ-I-550, Group 2".

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3.19, Table II, under Requirements, delete "MS17160-29" and substitute "MIL-C-24066/2"; delete "MS39062" and substitute "MIL-C-13516/1".

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3.24, line 3, delete and substitute: "MIL-STD-461A, Notice 1, III D."

3.25, delete entirely and substitute:

"3.25 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition as specified in 3.25.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

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When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.25.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II is used, the supplier shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66 - Air Pollution Control District, County of Los Angeles, CA."

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3.26, delete and substitute:

"3.26 Lifting attachments. Not less than four permanently affixed attachments that enable the heater to be lifted in its normal travel position shall be provided. Each attachment shall withstand, without permanent deformation to any part of the heater, a vertical static pull equal to 2.5 times the empty heater weight divided by the number of attachments. Attachments for multiple slings shall be located so that: (a) slings will converge at not more than 24 feet above the lowest extremity of the heater, and (b) not less than 1-inch clearance shall be maintained between the slings and the heater. Eye openings and clearance dimensions of the attachments shall conform to figure 4. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.27."

3.27, line 9, add: "Eye openings shall be in accordance with figure 4."

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3.31, line 3 after type A add: "color as specified (see 6.2)."

3.33, delete in its entirety.

3.35, delete in its entirety.

After 3.35, add:

"3.36 Rail transportability. Unless otherwise specified (see 6.2), the heater shall withstand rail impact when tested as specified in 4.6.2.13 without permanent deformation to any component. After this test the heater shall operate for a period of not less than 1 hour without malfunction or permanent deformation of any part of the heater."

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After 4.1.1, add:

"4.1.2 Acceptability criteria. Heaters which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations and tests in section 4 of this specification will be considered acceptable by the Government."

4.3.1, line 4, delete "3.15.21" and substitute "3.15.2.1".

4.3.3, line 2, delete "4.6.2.10" and substitute "4.6.2.13".

4.4, delete and substitute:

"4.4 Initial production inspection. When specified (see 3.3), one or more initial production heaters will be selected at random by the Government from the heaters being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.1, and 4.6.2.3 through 4.6.2.13 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production heater shall not exclude the remaining heaters from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests."

4.4.1, line 4, delete "heater" and substitute "heaters".

Page 32

4.6.1, number 114, delete "Chassis" and substitute "Trailer Components missing or".

4.6.1, add: "125. Servicing and adjustments not properly performed."

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4.6.2.3, line 2, delete "2 hours" and substitute "20 hours".

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Table III, Road Test, delete "55 mph" and substitute "45 mph".

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4.6.2.5 (b), line 1 and line 5, delete "50 mph" and substitute "45 mph".

Page 36

4.6.2.7, line 3, delete "19 feet" and substitute "24 feet".

4.6.2.7.1, line 3, delete "19 feet" and substitute "24 feet".

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After 4.6.2.11, add:

"4.6.2.12 Noise level measurement.

4.6.2.12.1 Representative duty station. Measure the sound level at eight evenly spaced points, 68 inches above ground level, on a circle of 15 foot radius from the geometric center of the equipment in accordance with MIL-STD-1474 for steady-state noise. The operating condition shall be no-load maximum governed speed.

4.6.2.12.2 Operator's station. Measure the sound level at the operator's station in accordance with MIL-STD-1474 for steady-state noise at the operating conditions of 4.6.2.3 (see 6.6).

4.6.2.12.3 Exterior spectator. Measure the exterior sound level in accordance with MIL-STD-1474 for exterior noise.

4.6.2.12.4 Failure criteria. Nonconformance to 3.5.2 shall constitute failure of this test.

4.6.2.13 Rail impact. Anchor the heater on a standard railway flatcar. Loading, blocking and securing shall be in accordance with section 6 of Association of American Railroads, Rules Governing the Loading of Department of Defense Material on Open Top Cars. Hump the car with the heater tied down, at a speed of not less than 8 mph against five empty boxcars with their brakes set. The heater shall face in the direction of the five cars. Repeat the procedure with the rear of the heater facing the five cars. The heater may be reversed by either reversing it to the proper direction on the flatcar or by reversing the direction of the flatcar. Once the test has begun, there will be no readjustment of load, or any reconditioning of the blocks, chocking material or tiedowns except to reverse direction of the heater. After each impact, the heater and tiedowns shall be inspected to determine permanent deformation to components of the heater or other evidence of possible failure. After completion of this test, a 1-hour operational checkout shall be performed to demonstrate

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proper operation of the heater and all controls. Weld failure or permanent deformation of the heater components or its tiedown attachments, or failure of the heater to operate after the test for not less than 1 hour without malfunction shall constitute failure of the test."

Page 37

4.7, delete and substitute:

"4.7 Inspection comparison. The Government may select heaters at any time during the contract production period and subject these heaters to the examination specified in 4.6.1 and the tests specified in 4.6.2.1, 4.6.2.2 and 4.6.2.3 through 4.6.2.13 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production heaters. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison heater to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison heaters and shall be cause for refusal by the Government to continue acceptance of production heaters until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on heaters previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all heaters accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer."

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6.2, delete and substitute:

"6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(d) When initial production is required, and number of initial production heaters to be furnished, when applicable (see 3.3).
(e) When the noise level shall be other than as specified (see 3.5.2).
(f) When a commercial engine is required (see 3.15.2).
(g) When wide-base tires are required (see 3.17.8.2).
(h) When standard military trailer components are required (see 3.19).
(i) When fungus and moisture resistance is not required (see 3.25).
(j) Color required (see 3.31).
(k) When rail impact test is not required (see 3.36).
(l) Level of preservation and packaging and level of packing required (see 5.1)."

After 6.4, add:

"6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of heaters. A suggested paragraph is as follows:

'Shipment of heater shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract.'

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6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each heater. When the noise level exceeds 85dB(A) when tested as specified in 4.6.2.12.2, the manuals should discuss the noise hazard per MIL-STD-1474, including the requirement for hearing protection, the noise level of the heater, the distance from the heater at which 85dB(A) will always be met, and what operational or system configuration will reduce the level to 85dB(A)."

After figure 3, add figure 4.

Custodians:

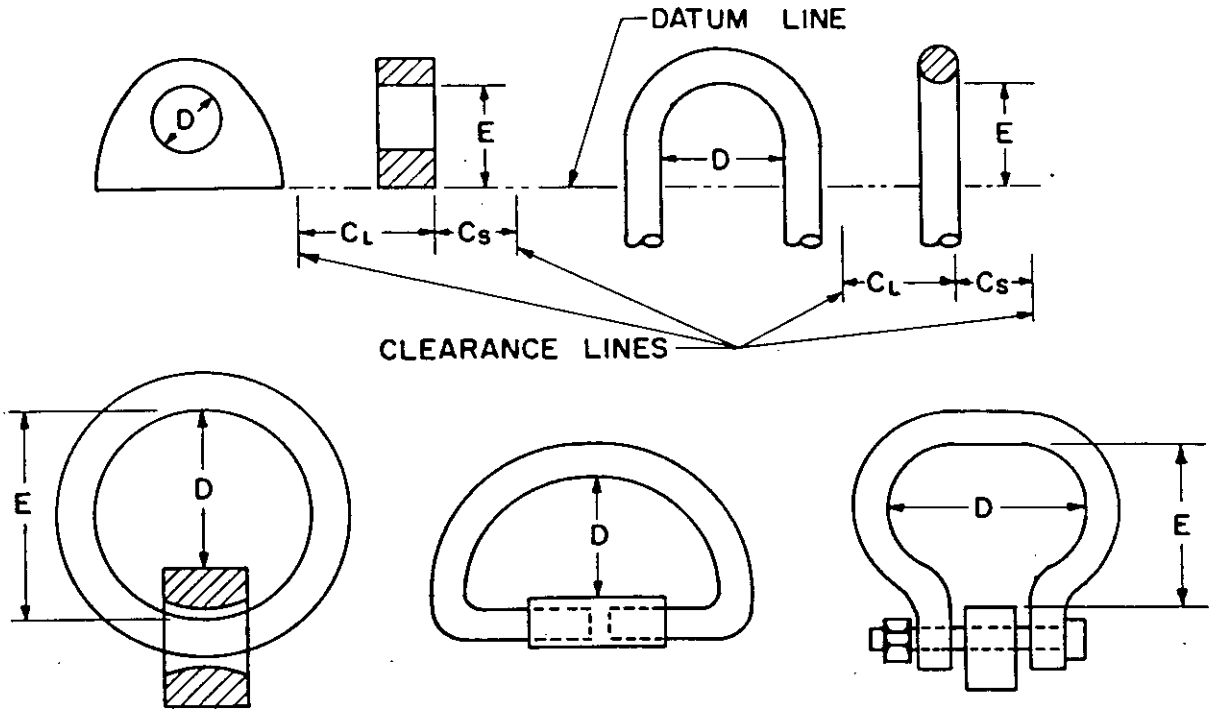
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DIMENSIONS IN INCHES						
Weight range of equipment (tons - 2,000 lbs per ton)	* C_L min.	* C_S min.	D max.	D min.	E max.	E min.
Up to 5	7	3	4	2	5	2-1/2
5 to 11	9	4	4	2-1/2	5	3
11 to 25	12	5	4	3	5	3-1/2
25 to 50	16	7	4-1/2	3-1/2	6	5
50 to 75	20	8	5	4	6	5-1/2

*There shall be no interference within C_L and C_S which could interfere with engaging a shackle and pin in the eye. Either side of the eye may be used as the datum from which to measure C_L and C_S .

NOTES:

1. Eye may be similar to any style shown.
2. Eye may be designed to swivel.

FIGURE 4. CLASS 1, 2, AND 3 OPENINGS AND CLEARANCE DIMENSIONS FOR EYES, TIEDOWNS, PROVISIONS AND SHACKLES ON TYPES I, II, III AND IV EQUIPMENT.

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